

The Algorithmic Payload Decision Engine

Instant mathematical evaluation of `sizes` and `srcset` in the Pre-parser (No CSS Layout required).

Environment Constraints

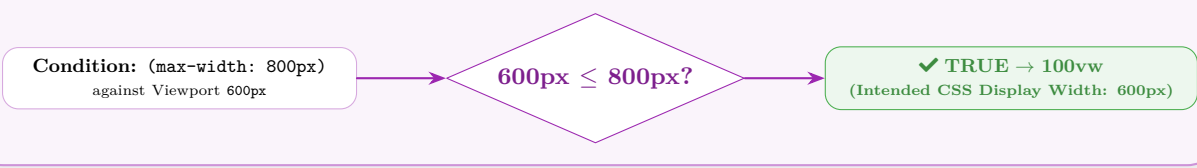


Browser Viewport: 600px
Device Pixel Ratio: 2.0x (Retina)

HTML Native Payload

```
index.html
<img class="hero"
  sizes="(max-width: 800px) 100vw, 50vw"
  srcset="img-400.jpg 400w,
    img-800.jpg 800w,
    img-1200.jpg 1200w,
    img-1600.jpg 1600w" />
```

Step 1: Evaluate sizes Media Conditions



Step 2: Hardware Pixel Translation

600px
CSS Width (100vw)

×

2.0
DPR (Retina)

=

1200
Required Pixels

Step 3: srcset Array Selection

Browser engine scans `srcset` for closest optimal match ≥ 1200.

img-400.jpg	400w	✗ Discarded (Too small)
img-800.jpg	800w	✗ Discarded (Too small)
img-1200.jpg	1200w	✓ Optimal Match Selected!
img-1600.jpg	1600w	— Ignored (Overkill / Wastes bandwidth)



INSTANT DISPATCH

The Pre-parser thread issues the network request for `img-1200.jpg` instantaneously—completely bypassing the CSS Layout calculation wait time.